Glasgow College, UESTC

Interim Report on UESTC4006P(BEng) Final Year Project

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Degree programme	Beng(Hons) Electronics and Electrical Engineering with Information Engineering (JEP with UESTC)
Academic year	Fourth Year

Placement Company (if appropriate)	
Working Title of Project	Research on Deep Learning based Fine Segmentation Technology of Retinal Blood Vessels in the Fundus
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Declaration of Originality and Submission Information	*I affirm that this submission is all my own work in accordance with the University of Glasgow Regulations and the School of Engineering requirements* YiChi Chen Signed (Student) :

1. Project Outline

1.1 Motivation and Background

Retinal vascular morphological features are critical biomarkers for diagnosing systemic diseases such as diabetic retinopathy and hypertension [1， 2]. While deep learning, particularly the nnU-Net framework, has set new benchmarks via automated self-configuration [3], accurately segmenting fine-grained terminal vessels remains a challenge. This project leverages the robust foundation of nnU-Net, incorporating attention mechanisms to enhance segmentation precision in complex retinal environments.

1.2 Problem Statement

Despite advancements, three critical bottlenecks persist in automated segmentation:

- Low Contrast: Terminal vessels often blend into the background, leading to significant missed detections [1].
- Topological Discontinuity: Downsampling in standard encoder-decoder structures causes "vessel breaking," where slender branches become fragmented [4， 5].

- Generalization Gap: Most models require manual tuning for different datasets, limiting clinical deployment [3].

To address these, this research proposes an enhanced nnU-Net integrating Attention Gates (AGs) to suppress noise and prioritize small-scale vascular features [6].

1.3 Research Objectives and Methodology

This research adopts a phased technical approach to overcome the structural and signal-level limitations in retinal vessel segmentation. The integrated objectives and corresponding methodologies are as follows:

- **O1: Contrast Enhancement via CLAHE:** Integrate Contrast Limited Adaptive Histogram Equalization (CLAHE) into the preprocessing pipeline to normalize illumination and amplify the visibility of micro-vessels in low-contrast regions, ensuring better feature capture for the initial convolutional layers.
- **O2: Hard-Pixel Mining with Focal Loss:** Address extreme class imbalance by replacing the standard Dice loss with Focal Loss. This forces the model to prioritize "hard-to-segment" pixels, such as thin vessel boundaries and misclassified samples, rather than being dominated by the easily classified background.
- **O3: Architectural Optimization with Attention Gates:** Modify the nnU-Net backbone by incorporating Attention Gates (AGs) into the skip connections [6]. This allows the decoder to selectively fuse spatial information, effectively suppressing background noise while preserving the topological connectivity of slender vascular branches.
- **O4: Performance Benchmarking:** Systematically evaluate the enhanced model against state-of-the-art (SOTA) methods using the DRIVE and STARE datasets to verify improvements in segmentation precision and robustness.

1.4 Proposed Research Methodology

To quantitatively assess the model's segmentation performance and diagnostic reliability, the following standard metrics are utilized:

- Dice Similarity Coefficient (DSC) for overall overlap accuracy.
- Sensitivity (Recall) to evaluate the detection of fine-grained terminal branches.
- F1-Score and AUC-ROC to measure the trade-off between precision and robustness.
- Accuracy (Acc) for general pixel-wise classification performance

2.Work done so far

2.1Experimental Environment and Dataset Configuration

2.1.1 Dataset and Training Strategy

The experimental process was divided into two strategic phases:

- Phase 1: Generalization Testing (Exp 1–3): I used external datasets (STARE and CHASE_DB1, etc) with a fixed split for training. These models were evaluated on the DRIVE test set to measure cross-dataset robustness [7-10].
- Phase 2: In-domain Optimization (Exp 4–5): I utilized the DRIVE dataset with 5-fold cross-validation. The data was partitioned into five subsets, iteratively training on four and validating on one to minimize partitioning bias.

2.1.2 Hardware and Software Specifications

Category	Specifications
Backbone Framework	nnU-Net V2 (Self-configuring pipeline)
Software Environment	Python 3.10, PyTorch 2.9.0, CUDA 11.7
Hardware Resources	NVIDIA RTX 4060 (8GB VRAM)
Base Learning Rate	0.01
LR Schedulers	Poly (Exp 1–4); Cosine Annealing (Exp 5)
Optimization	Batch sizes dynamically optimized for 8GB VRAM

2.2 Experiment 1: Baseline Implementation and Data Engineering

The first phase of the technical implementation focused on establishing a robust performance baseline using the nnU-Net (no-new-U-Net) framework. This experiment served as the control group to evaluate the framework's "out-of-the-box" capability in handling retinal vessel segmentation without domain-specific modifications to the loss function or architecture.

2.2.1 Data Standardization and Multi-Channel Strategy

To meet nnU-Net's input requirements, raw images were converted to NIfTI format using a multi-channel decomposition strategy that treats RGB channels as independent modalities. This approach forces the model to learn specific spectral features, particularly from the high-contrast green channel, while standardized spatial calibration ensures geometric consistency across all heterogeneous datasets.

2.2.2 Training Configuration and Baseline Analysis

This experiment aimed to evaluate nnU-Net's "out-of-the-box" performance on retinal vessel segmentation.

- Dataset and Training: I aggregated data from STARE, CHASE_DB1, and HRF to form a training pool of 802 images with a standardized resolution of 512 x 512 pixels. Training utilized a composite of Dice and Cross-Entropy loss over 150 epochs.
- Quantitative Results: The model achieved convergence at the 69th epoch and recorded a Dice coefficient of 0.8617 on the DRIVE test set.
- Limitation Analysis: Qualitative inspection revealed that while the model effectively segmented major vessels, it frequently missed fine-grained terminal branches due to low local contrast. This failure of standard Z-score normalization and Dice loss to prioritize micro-structures directly motivated the implementation of CLAHE and Focal Loss in subsequent experiments.

2.3 Experiment 2: Targeted Enhancement via Green-Channel CLAHE

The second experiment aimed to overcome the physical limitations of fundus images—specifically non-uniform illumination and low local contrast—which were identified as primary causes of terminal vessel omission in Experiment 1.

2.3.1 Motivation and Analysis of Training Dynamics

The green channel was prioritized for enhancement as it exhibits the highest contrast between hemoglobin-filled blood vessels and retinal tissue. Integrating CLAHE on this modality acted as a "feature catalyst" [11-12], improving the initial Dice score by 5.39% and accelerating convergence by 13% (60 epochs vs. 69 in Exp 1).

As shown in Table 3, while CLAHE lowered optimization difficulty early in training, the final Dice value remained nearly identical to the baseline. This empirical plateau confirms that enhancing the input signal alone is insufficient to bypass the structural limitations of the standard U-Net, justifying the transition to architectural modifications in Phase 2.

Table 3: Training Progress Comparison (Exp 1 vs. Exp 2)

Feature	Experiment 1 (Baseline)	Experiment 2 (Green-Channel CLAHE)
Initial Dice Coefficient	0.7128	0.7512 (+5.39%)
Convergence Epoch	69	60 (13% faster)
Final Test Dice (DRIVE)	0.8617	0.8619

2.3.2 Mathematical Foundation of CLAHE

To mitigate performance saturation, CLAHE operates on localized tiles to enhance regional contrast rather than applying global equalization.

The algorithm introduces a clip limit to prevent noise over-amplification in homogeneous areas by clipping histogram bins that exceed a predefined threshold and redistributing excess pixels uniformly. This mechanism sharpens terminal vessel edges while preserving a noise-free background, providing stable structural signals for the nnU-Net encoder.

2.4 Experiment 3: Optimization via Focal Loss for Hard-Pixel Mining

In Experiment 3, I focused on addressing the inherent class imbalance and the "hard-pixel" problem in retinal vessel segmentation. While the previous experiments successfully enhanced the input signal via CLAHE, the model still lacked the necessary focus on fine-grained terminal vessels and blurry vascular boundaries. To rectify this, I maintained the Green-Channel CLAHE preprocessing and transitioned the optimization target to Focal Loss.

2.4.1 Theoretical Motivation: Class Imbalance and "Hard" Pixel Focus

Retinal images exhibit extreme class imbalance, with vessel pixels often occupying less than 10% of the total area. Standard loss functions are frequently dominated by "easy" background pixels, causing the model to neglect difficult-to-classify micro-structures. By introducing a modulating factor $(1 - p_t)^{\gamma}$ to the standard cross-entropy, Focal Loss down-weights the loss contribution from easy examples [13-15]. This forces the network to concentrate its learning capacity on "hard" pixels where the prediction probability is low, effectively mining for vascular features in low-contrast regions that were previously treated as background noise.

2.4.2 Training Dynamics and Convergence Performance

The training was conducted for 150 epochs. I observed that the combination of targeted data enhancement and a focused loss function significantly streamlined the optimization process:

- Convergence and Loss: The model reached its optimal state at Epoch 53, achieving a minimum Focal Loss value of 0.0246. This marks a 23% improvement in convergence speed compared to the baseline (69 epochs), suggesting that the targeted objective function allowed the gradient descent to navigate the feature space more efficiently.
- Metric Stabilization: At the point of convergence, the model recorded a validation Dice coefficient of 0.8584, reflecting a stable learning curve without significant overfitting.

2.4.3 Quantitative Evaluation and Technical Trade-offs

The performance metrics for Experiment 3 are detailed in the table below:

Table 4: Detailed Performance Metrics for Experiment 3

Metric	Value
IoU (Vessel)	0.6410
mIoU	0.8023
Recall	0.7255
Precision	0.8489
Accuracy	0.9659
F1-score	0.7802

A rigorous comparison between Experiment 2 and Experiment 3 reveals that the integration of Focal Loss led to a slight decline across most segmentation metrics. Specifically, the Precision dropped from 0.8497 (Exp 2) to 0.8489 (Exp 3), and the F1-score decreased from 0.7827 to 0.7802.

This performance regression indicates that for the DRIVE and STARE datasets, the pixel-wise weighting of Focal Loss does not yield a direct accuracy boost over the standard Dice-based optimization within the current U-Net architecture. However, Experiment 3 demonstrated superior training efficiency, reaching convergence at Epoch 53—the fastest among all preliminary trials. More importantly, these results serve as critical evidence that the performance of the model has reached a plateau under the standard encoder-decoder structure. This justifies the transition to the next research phase, where architectural modifications (LPT and AFFM) are introduced to overcome these fundamental structural limitations.

2.5 Summary of Preliminary Experiments and Comparative Analysis

To evaluate the effectiveness of my work, I compared my first three experiments against prior results from Senior Liu Zhishuai. All models were trained for 150 epochs on the same dataset.

Table 5: Performance Comparison of Segmentation Methods

Segmentation Method	IoU (Vessel)	mIoU	Recall	Precision	Accuracy	F1-score
Prior Work (Senior Liu)						
Hrnet	0.4686	0.6995	0.6890	0.5943	0.9344	0.6382
Deeplabv3+	0.5764	0.7601	0.8388	0.6483	0.9477	0.7313
UNet	0.6441	0.8026	0.7735	0.7937	0.9637	0.7835
CLAHE + Improved UNet	0.6560	0.8089	0.8002	0.7845	0.9644	0.7923
Current Work (I)						

Segmentation Method	IoU (Vessel)	mIoU	Recall	Precision	Accuracy	F1-score
Exp 1: nnU-Net Baseline	0.6574	0.8107	0.7688	0.8219	0.9664	0.7924
Exp 2: CLAHE + nnU-Net	0.6441	0.8041	0.7282	0.8497	0.9662	0.7827
Exp 3: CLAHE + nnU-Net + FL	0.6410	0.8023	0.7255	0.8489	0.9659	0.7802

Key Insights:

- Framework Superiority: My nnU-Net baseline (Exp 1) outperformed all of Senior Liu's models in almost every key metric, including IoU (0.6574) and Accuracy (0.9664). This confirms that the automated "data fingerprinting" and self-configuring pipeline of nnU-Net provide a significantly more robust foundation for retinal vessel segmentation than manually tuned traditional architectures.
- Preprocessing Impact: The introduction of Green-Channel CLAHE (Exp 2) achieved the highest Precision (0.8497) among all tested methods. This verifies that enhancing the input signal at the source is the most effective way to reduce false-positive noise and improve boundary selectivity.
- The Structural Performance Plateau: The transition to Focal Loss (Exp 3) resulted in the fastest convergence (Epoch 53) but failed to yield a breakthrough in segmentation accuracy, with Precision dropping slightly to 0.8489 compared to Experiment 2. This performance regression is a critical indicator that the current encoder-decoder architecture has reached its physical limit.
- Logical Pivot to Phase 2: While pixel-level clarity was improved in the preliminary phase, topological discontinuities in terminal branches persist. This provides a clear justification for Experiment 4&5, where I would transition from purely optimizing "signals" (preprocessing/loss) to modifying the architecture itself through Attention Gates and hybrid CNN-Transformer modules

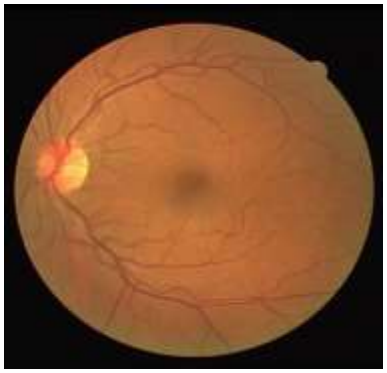


Figure 1 Original image



Figure 2 CLAHE enhancement image

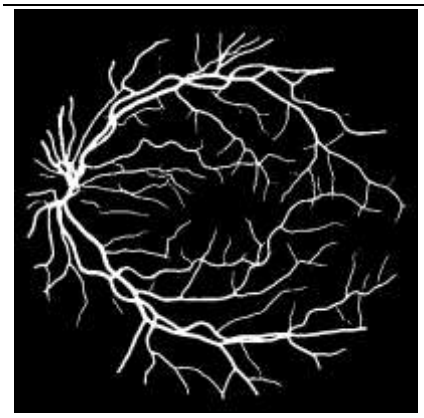


Figure 3 Experiment1 output

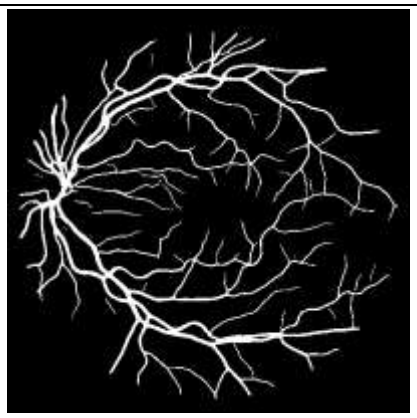


Figure 4 Experiment2 output

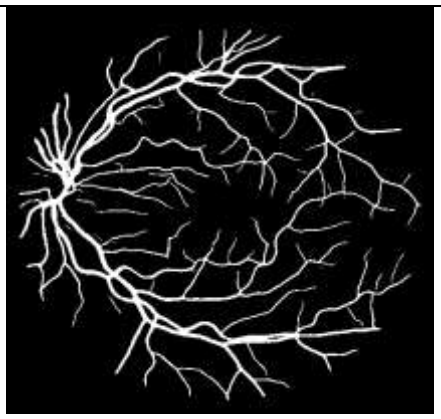


Figure 5 Experiment3 output

2.6 Experiments 4 & 5: Architectural Migration and Framework Compatibility Analysis

Following the preliminary optimizations, I shifted the research focus toward the architectural principles of PA-Net, a state-of-the-art hybrid CNN-Transformer model. Experiments 4 and 5 were conducted as a controlled group to evaluate whether PA-Net's domain-specific preprocessing and core modules could be successfully migrated into the nnU-Net framework.

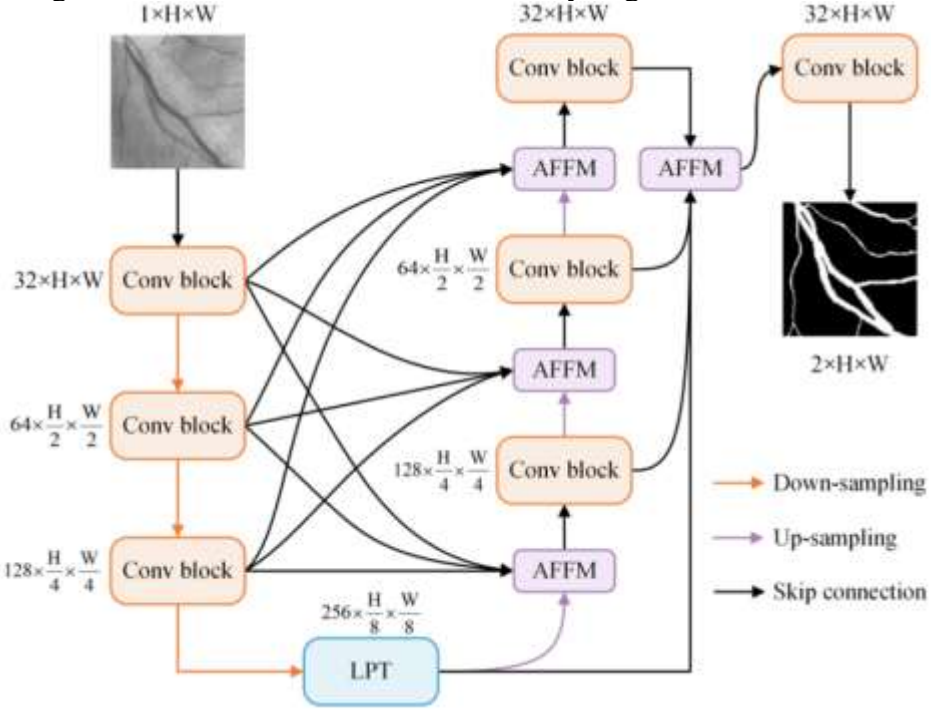


Figure 6: Overview of the PA-Net hybrid architecture[4]

2.6.1 Experiment 4: Standardized Preprocessing and 5-Fold Cross-Validation

To establish a fair benchmark against the PA-Net paper [4], I implemented their four-step preprocessing pipeline—comprising Graying, Standardization, CLAHE, and Gamma Transform—on the DRIVE dataset. I transitioned to a 5-fold cross-validation strategy and developed a custom

batch evaluation script for standardized metric calculation. Notably, the nnU-Net baseline recorded an AUC of 0.8739, which is significantly lower than the 0.9833 reported in the PA-Net paper. This discrepancy suggests that many SOTA results may rely on highly specific training conditions that are difficult to replicate within generalized frameworks like nnU-Net.

2.6.2 Experiment 5: Integrating AFFM and LPT into nnU-Net

In Experiment 5, I modified the nnU-Net architecture by integrating PA-Net's core components and switched the optimizer to Adam:

- LPT (Lightweight Parallel Transformer): I embedded the LPT module into the bottleneck layer to capture long-range dependencies and prevent slender vessels from breaking The LPT follows the definition:

$$\mathrm{LPT}(x) = \mathrm{CFFN}\left(\mathrm{EPA}\left(\mathrm{LFAU}(x)\right)\right)$$

where LFAU (Local Feature Accepting Unit) embeds positional information and EPA (Efficient Parallel Attention) utilizes parallel gated axial-attention to minimize computational cost.

- AFFM (Adaptive Vascular Feature Fusion Module): I replaced the standard skip connections with AFFM to mitigate information loss during downsampling. Unlike simple concatenation, AFFM adaptively fuses semantic information from different levels:

$$\mathrm{F}_{\mathrm{output}} = [(C_1 \times A_1 + C_2 \times A_2 + C_3 \times A_3), F_0]$$

Here, A1, A2, and A3 are spatial attention weights generated via a Softmax function to guide the fusion of feature maps from multiple depths.

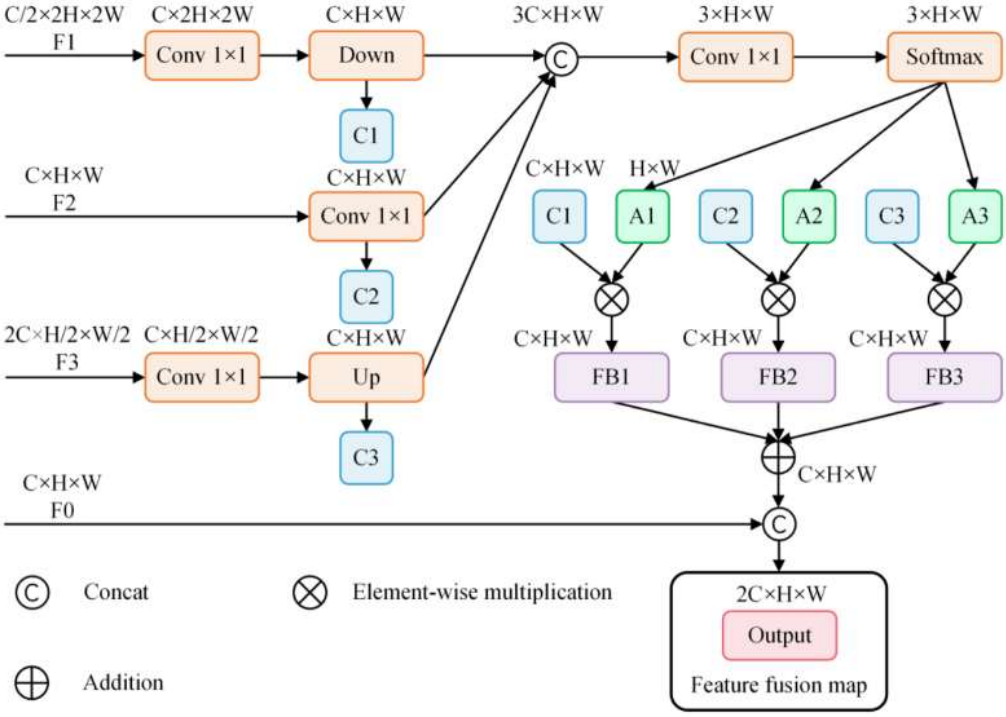


Figure 7 Mechanism of the Adaptive Vascular Feature Fusion Module (AFFM)[4]

2.6.3 Results and Comparative Analysis

The comparative results between the vanilla nnU-Net and the modified version are summarized in Table 6.

Table 6: Performance Metrics Comparison (Mean ± Std)

Metric	nnU-Net v2	nnU-Net + LPT + AFFM	PA-Net (Paper)	U-Net (Paper)
SE (Sensitivity)	0.7627 ± 0.0446	0.7556 ± 0.0454	0.82841	0.77002
SP (Specificity)	0.9852 ± 0.0050	0.9858 ± 0.0050	0.98073	0.98294
ACC (Accuracy)	0.9659 ± 0.0045	0.9658 ± 0.0047	0.95825	0.95486
AUC	0.8739 ± 0.0210	0.8707 ± 0.0214	0.98337	0.97898
PR (Precision)	0.8238 ± 0.0646	0.8283 ± 0.0662	0.85049	0.87301

2.6.4 Conflict Mechanism: Deep Supervision vs. AFFM

The drop in Sensitivity (SE) and AUC following architectural integration likely stems from a structural conflict between the Adaptive Vascular Feature Fusion Module (AFFM) and nnU-Net's Deep Supervision mechanism. While Deep Supervision requires clean, level-specific gradients for auxiliary loss calculation at every decoder level, the AFFM forces the fusion of features from multiple depths (F_1, F_2, F_3) into a single path. This disrupts the hierarchical gradient flow, leading to mixed signals and "negative optimization". Upcoming ablation studies will independently evaluate the Lightweight Parallel Transformer (LPT) and AFFM to isolate these effects.

3.Challenges and Solutions

The implementation of this project involved navigating several technical hurdles, ranging from framework deployment to coordinate system calibration. Overcoming these obstacles required a deep understanding of medical imaging standards and the inner workings of the nnU-Net pipeline.

3.1 Framework Deployment and Environment Configuration

As a researcher initially unfamiliar with the PyTorch ecosystem, the deployment of nnU-Net v2 presented a steep learning curve. The framework possesses a complex dependency tree, which led to significant versioning conflicts between batchgenerators, CUDA drivers, and the core PyTorch library.

- Solution: I resolved these "versioning hell" issues by building isolated Conda environments and manually pinning library versions to ensure compatibility with Python 3.10 and CUDA

11.7. This established a stable foundation for the self-configuring pipeline to initialize without runtime errors.

3.2 Dimensionality and Coordinate Calibration (The 180-Degree Bug)

A primary challenge involved the transition from 2D images (DRIVE/STARE) to the 3D-native NIfTI format required by nnU-Net.

- Pseudo-3D Conversion: I initially struggled with the framework's requirement for volumetric data. I eventually developed a script to "trick" the 3D engine by explicitly setting the depth dimension to 1 and normalizing the affine matrix to (1.0, 1.0, 1.0).
- Orientation Correction: During initial testing, I discovered that the model's output masks were consistently rotated 180 degrees relative to the Ground Truth labels. This resulted in deceptively low initial metrics. I traced this error to an axis-ordering mismatch during the JPG-to-NIfTI conversion. By recalibrating the affine transformation matrix and verifying the spatial headers, I corrected the orientation, restoring the metrics to their true values.

3.3 Training Efficiency and Feedback Loops

- Time Constraints: Given the high resolution of retinal images, each training session (150 epochs) on my local hardware requires approximately six hours. This long feedback loop limited the frequency of architectural iterations.
- Solution: To maintain progress, I automated the evaluation process using batch scripts. For the final high-complexity runs, I have prepared a transition to a high-performance server to facilitate faster ablation cycles.

4. Conclusions and Future Work

4.1 Conclusions from Initial Work

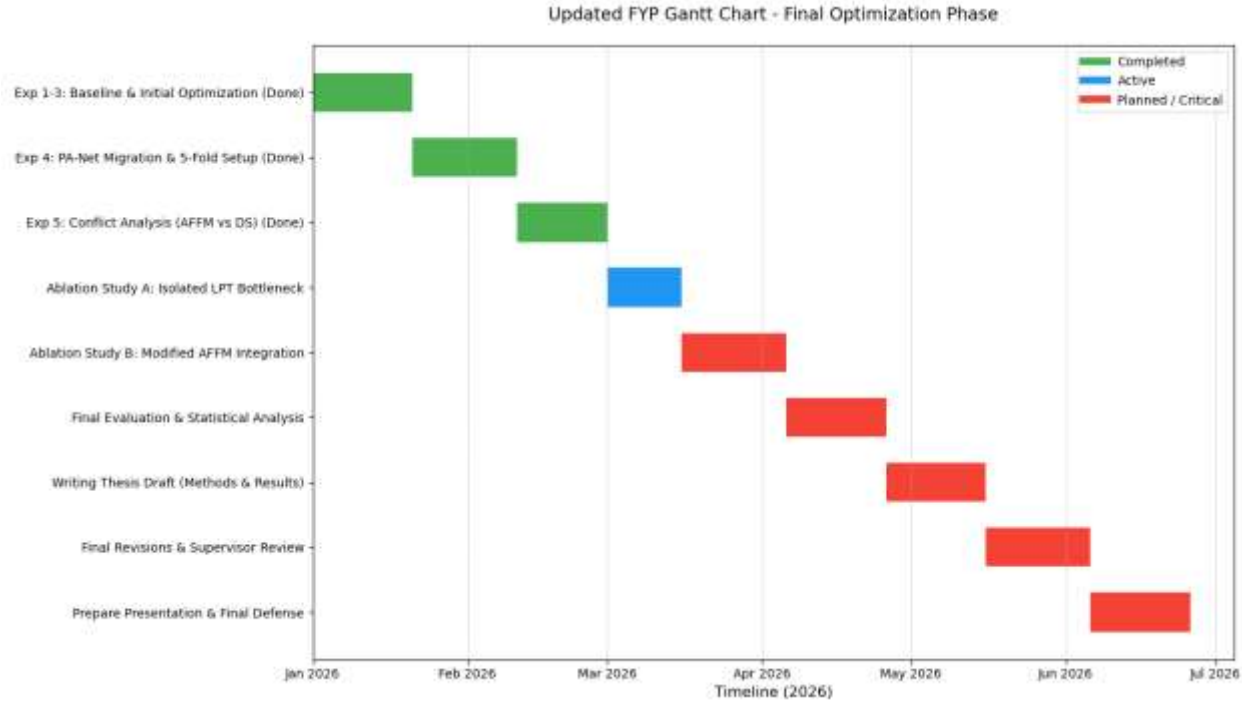
The first phase of this project has yielded several critical insights:

1. nnU-Net Baseline Strength: The automated configuration of nnU-Net provides a remarkably high-performance floor, achieving a Dice of 0.8617 without manual tuning.
2. Strategic Trade-offs: Integrating CLAHE reduced the convergence time by 13% (from 69 to 60 epochs). While Focal Loss achieved the fastest convergence at 53 epochs, it resulted in a performance plateau where Precision (0.8489) was slightly lower than the CLAHE-only model (0.8497). This proves that loss-weighting alone cannot overcome the structural bottlenecks of the encoder-decoder architecture.
3. Architectural Compatibility: The "negative optimization" observed when adding PA-Net modules suggests a structural conflict where the AFFM disrupts the hierarchical gradient flow required by nnU-Net’s Deep Supervision mechanism.

4.2 Future Work and Timeline

Moving forward into the final semester of 2026, my research will focus on isolating the benefits of the PA-Net modules:

- Ablation Study A (LPT Focus): I will test the Lightweight Parallel Transformer (LPT) in the bottleneck layer alone to evaluate its impact on capturing long-range dependencies without the interference of skip-connection modifications.
- Ablation Study B (Modified AFFM): I plan to decouple the AFFM fusion from the Deep Supervision layers to see if the model can benefit from multi-level feature fusion without losing gradient clarity.
- Thesis Finalization: The project is scheduled to conclude in June 2026 with the final defence and submission of the research findings.



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